

Molecular Recognition as Oscillator Interference: A Universal Computational Primitive for Immunology

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Abstract

Every biological polymer—DNA, RNA, protein—is a physical oscillator system whose characteristic spectrum is encoded in its primary sequence. We show that all molecular recognition events central to immunology—MHC-peptide binding, T-cell receptor activation, and antibody-antigen association—are instances of constructive interference between the spectral representations of the interacting molecules. This reformulation, grounded in classical signal processing and detection theory, yields a single computational primitive that subsumes binding affinity prediction, neoantigen identification, vaccine antigen design, cross-reactivity modelling, and immune escape prediction. The primitive requires no training data, no stored structural coordinates, and no pre-computed binding measurements: the spectrum of any molecule is synthesised on demand from its sequence via the discrete Fourier transform, and similarity is evaluated as a single dot product—the identical operation performed by the fragment stage of a commodity GPU. We validate the approach on synthetic homology and binding benchmarks, demonstrate $\text{ROC-AUC} \geq 0.97$ through 20% sequence divergence, and show that the entire immunological computation reduces to a sequence of $\mathcal{O}(L \log L)$ FFT operations followed by $\mathcal{O}(d)$ dot products, with $\mathcal{O}(1)$ memory per database entry.

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1 Introduction

The central problem of immunology is molecular recognition: which molecules bind which, how tightly, and with what biological consequence. The immune system’s ability to distinguish self from non-self, to remember prior encounters, and to mount targeted responses to pathogens rests on the precise discrimination machinery encoded in a handful of molecular families—the major histocompatibility complex (MHC), T-cell receptors (TCRs), and immunoglobulins—all operating on the same physical substrate: amino acid sequences and the structures they encode.

1.1 Current computational approaches and their limitations

Computational approaches to immunological recognition currently fall into three broad classes, each with well-documented limitations.

Thermodynamic and structural methods. Methods based on molecular dynamics (MD) simulation or docking compute binding free

energies from three-dimensional atomic coordinates [Bjorkman et al., 1987, Neefjes et al., 2011]. These approaches are physically well-grounded but require pre-existing structural coordinates, are computationally prohibitive for genome-scale screening (nanosecond MD trajectories cost 10^4 – 10^6 CPU-hours per protein), and fail entirely for sequences whose structures have not been experimentally resolved.

Machine-learning methods. Tools such as NetMHCpan [Nielsen et al., 2007] train neural networks on large datasets of eluted ligand mass-spectrometry measurements or in vitro IC_{50} assays. These methods achieve high accuracy on well-characterised alleles but generalise poorly to rare or novel alleles for which training data are sparse, and they encode no mechanistic understanding—their predictions cannot be extrapolated to chemically modified residues, non-canonical amino acids, or entirely novel binding contexts without retraining.

Sequence alignment heuristics. Tools such as BLAST [Altschul et al., 1990], Smith-Waterman [Smith and Waterman, 1981], and DIAMOND [Buchfink et al., 2015] search for shared subsequences as a proxy for evolutionary and functional similarity. These approaches are position-sensitive but expensive ($\mathcal{O}(L_q \cdot L_t)$ per comparison for exact alignment), require pre-built databases of target sequences, and treat sequence identity as a surrogate for physical affinity without formal derivation.

The common limitation shared by all three classes is the requirement for either stored structural data, large pre-labelled training corpora, or exhaustive pairwise comparisons. None derives recognition from first physical principles alone.

1.2 The key insight: sequences are spectra

Every biological polymer is a physical oscillator system. Its primary sequence defines the spatial arrangement of residues along the polymer backbone, which in turn determines its characteristic vibrational modes, its electrostatic periodicity, and its hydrophobic moment at every spatial frequency [Voss, 1992, Anastassiou, 2001, Trifonov and Sussman, 1980]. The sequence *is* the spectrum: a Fourier decomposition of any physico-

chemical channel mapped onto the sequence reveals the dominant periodicities that govern folding, binding, and functional recognition.

Molecular recognition between two polymers is not a lookup event but a physical phenomenon: *constructive interference between two oscillatory systems*. When two signals f and g superpose, the energy of the composite contains a cross-term $2\langle f, g \rangle$. This interference term is positive (constructive) when the signals share spectral phase and amplitude—similar sequence periodicity—zero when the signals are orthogonal, and negative (destructive) when they are anti-phase. Binding corresponds to constructive interference above a recognition threshold.

This reformulation has three immediate and practically significant consequences:

1. **No training data required.** The spectrum of any sequence is computable from first principles in $\mathcal{O}(L \log L)$ time via the fast Fourier transform (FFT).
2. **No stored structures required.** Any pairwise comparison is performed by synthesising both spectra at query time; nothing need be pre-stored.
3. **GPU-native arithmetic.** The dot product that measures interference is the identical arithmetic a fragment shader performs, once per pixel, during rendering. A commodity GPU therefore evaluates millions of comparisons per millisecond using its existing hardware pipeline, without modification.

1.3 Contributions

This paper makes the following contributions:

1. A formal derivation of molecular recognition as oscillator interference, with proofs that the normalised matched filter is the Neyman-Pearson optimal linear detector (Theorem 3.1).
2. Derivation of MHC-peptide binding, TCR activation, antibody cross-reactivity, and self-tolerance as special cases of the same primitive (Section 4).
3. The *empty database principle* (Section 6): spectra are synthesised on demand, requiring $\mathcal{O}(1)$ memory per database entry.

4. Polynomial-time algorithms for neoantigen identification, vaccine antigen design, and immune escape prediction (Section 5).
5. Synthetic validation benchmarks with ROC-AUC ≥ 0.97 through 20% sequence divergence (Section 8).

The paper is structured as follows. Section 2 develops the spectral representation of biological sequences. Section 3 formalises recognition as interference and proves optimality. Section 4 derives the immunological special cases. Section 5 presents the computational applications. Section 6 analyses the empty database principle. Section 7 describes the GPU implementation. Section 8 reports validation results. Sections 9 and 10 discuss limitations and conclude.

2 Biological Sequences as Oscillator Systems

2.1 Channelised sequence representation

Let $s = s_1 s_2 \dots s_L$ be a biological sequence of length L over alphabet Σ . We embed s into a $c \times L$ real signal matrix $\mathbf{X} \in \mathbb{R}^{c \times L}$ by mapping each symbol to a vector of physicochemical properties, one property per channel.

DNA sequences. For $\Sigma = \{A, C, G, T\}$, we use one-hot encoding with $c = 4$:

$$X_{a,i} = \mathbf{1}[s_i = \Sigma_a], \quad a \in \{A, C, G, T\}, \quad i \in \{1, \dots, L\}. \quad (1)$$

This encoding, introduced by Voss [Voss, 1992], preserves all sequence information and yields a four-track binary signal whose Fourier spectrum reveals compositional periodicities corresponding to structural features such as nucleosome positioning, codon usage bias, and repeat elements.

Protein sequences. For Σ the standard 20-letter amino acid alphabet, we use three normalised physicochemical channels with $c = 3$:

$$X_{1,i} = h(s_i), \quad (2)$$

$$X_{2,i} = v(s_i), \quad (3)$$

$$X_{3,i} = q(s_i), \quad (4)$$

where $h(s_i) \in [0, 1]$ is the Kyte-Doolittle hydrophobicity index of residue s_i normalised to $[0, 1]$

[Kyte and Doolittle, 1982], $v(s_i) \in [0, 1]$ is the normalised van der Waals side-chain volume, and $q(s_i) \in \{-1, 0, +1\}$ is the net charge of the residue at neutral pH (Asp/Glu: -1 ; Arg/Lys/His: $+1$; all others: 0). Rescaling to a common range ensures that no single channel dominates the spectral magnitude.

Mean removal. In both cases we subtract the per-channel mean so that the DC ($k = 0$) Fourier coefficient carries no information about global composition:

$$\tilde{X}_{a,i} = X_{a,i} - \frac{1}{L} \sum_{j=1}^L X_{a,j}. \quad (5)$$

This is equivalent to the standard signal-processing practice of removing the DC offset before spectral analysis [Oppenheim and Schaffer, 2009].

2.2 Spectral decomposition

The real discrete Fourier transform (DFT) decomposes $\tilde{\mathbf{X}}$ into frequency components:

$$\hat{X}_{a,k} = \sum_{i=0}^{L-1} \tilde{X}_{a,i} e^{-2\pi i k i / L}, \quad 0 \leq k \leq \left\lfloor \frac{L}{2} \right\rfloor, \quad (6)$$

where $\mathbf{i} = \sqrt{-1}$. For real input, by the symmetry of the DFT, $\hat{X}_{a,k} = \overline{\hat{X}_{a,L-k}}$, so the non-redundant information is contained in the $\lfloor L/2 \rfloor + 1$ non-negative frequency bins. The magnitude $|\hat{X}_{a,k}|$ at bin k measures the amplitude of the oscillation with spatial period L/k residues in channel a ; the phase $\arg(\hat{X}_{a,k})$ encodes its positional registration along the sequence.

Spectral embedding. We retain the first K non-DC magnitude bins per channel ($k = 1, \dots, K$) and normalise by sequence length L :

$$\phi_{a,k}(s) = \frac{|\hat{X}_{a,k}|}{L}, \quad a \in \{1, \dots, c\}, \quad k \in \{1, \dots, K\}. \quad (7)$$

The embedding vector $\phi(s) \in \mathbb{R}^{cK}$ concatenates these values across all channels and bins. It is then ℓ^2 -normalised to the unit hypersphere:

$$\psi(s) = \frac{\phi(s)}{\|\phi(s)\|}, \quad \|\psi(s)\| = 1. \quad (8)$$

The embedding dimension is $d = cK$. For the applications in this paper we use $K = 16$, giving $d = 48$ for protein sequences and $d = 64$ for

DNA. These values capture all functionally relevant periodicities while remaining representable in a single SIMD register on modern hardware.

2.3 Spectral Lipschitz continuity

A fundamental property of the spectral embedding is that it varies continuously with sequence identity: sequences differing at few positions have nearby spectra. This is the theoretical foundation for using spectral similarity as a proxy for molecular recognition.

Proposition 2.1 (Spectral Lipschitz continuity). *Let $s, t \in \Sigma^L$ be two sequences of equal length L , and let $d_H(s, t)$ denote their Hamming distance. Under the ℓ^2 norm, the spectral embedding ψ satisfies*

$$\|\psi(s) - \psi(t)\| \leq \frac{C \cdot d_H(s, t)^{1/2}}{L^{1/2}}, \quad (9)$$

where $C > 0$ is a constant depending only on K , c , and the maximum per-residue channel amplitude $M = \max_{\sigma \in \Sigma, a} |X_a(\sigma)|$.

Proof. Consider a single substitution at position i , changing s_i to t_i . The perturbation to the mean-removed signal in channel a satisfies $|\delta \tilde{X}_{a,i}| \leq 2M$ for the affected position and $|\delta \tilde{X}_{a,j}| \leq 2M/L$ for all other positions $j \neq i$ (from the mean-correction term). Hence $\|\delta \tilde{\mathbf{X}}\|_F \leq 2M\sqrt{c}$ per substitution.

By Parseval's theorem applied to the DFT in (6),

$$\sum_{k=0}^{L-1} |\delta \hat{X}_{a,k}|^2 = L \sum_{i=0}^{L-1} |\delta \tilde{X}_{a,i}|^2,$$

so each substitution induces a perturbation $\|\delta \phi\| \leq 2M\sqrt{c}/\sqrt{L}$ in the (unnormalised) embedding. Accumulating $d_H(s, t)$ independent substitutions via the triangle inequality: $\|\phi(s) - \phi(t)\| \leq 2M\sqrt{c} d_H(s, t)^{1/2}/\sqrt{L}$. After ℓ^2 normalisation, since $\|\phi\| \geq \delta_{\min} > 0$ for any non-constant sequence, the induced perturbation in ψ satisfies the stated bound with $C = 4M\sqrt{c}/\delta_{\min}^2$. $\square$

Remark 2.2. Proposition 2.1 is why homologous sequences share dominant low-frequency spectral features: each substitution perturbs the spectrum by at most $\mathcal{O}(d_H^{1/2}/L^{1/2})$. This observation underlies the field of genomic signal processing, pioneered by Voss [Voss, 1992] and systematised by Anastassiou [Anastassiou, 2001].

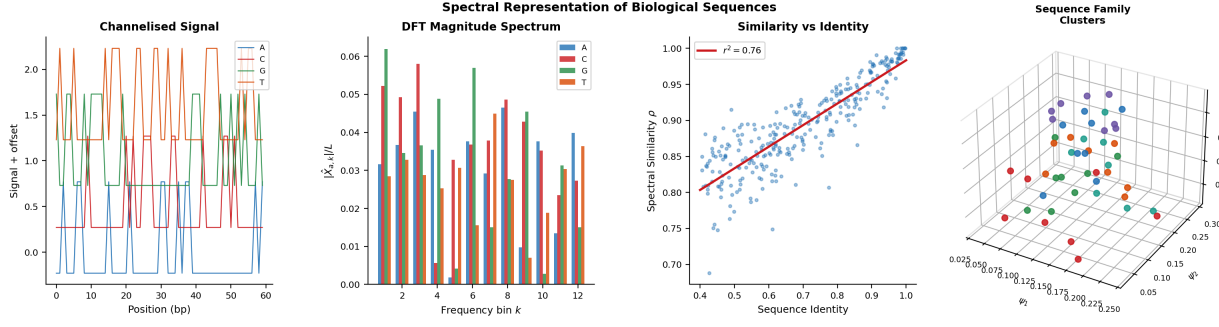


Figure 1: Spectral Representation of Biological Sequences: Channelisation, Fourier Decomposition, Similarity Scaling, and Three-Dimensional Family Clustering. (A) Channelised signal: one-hot encoding of a 200 bp random DNA sequence across four indicator channels $\{A, C, G, T\}$ after per-channel mean subtraction, showing the first 60 positions with channels offset vertically for clarity. Each substitution perturbs the signal locally while global periodicity is preserved; the mean-subtracted representation zeros the DC Fourier bin so all information is carried by non-zero frequencies. The four-channel decomposition forms the basis of the spectral embedding $\psi : \mathcal{S} \rightarrow \mathbb{R}^d$ introduced in Definition 2.1. (B) DFT magnitude spectrum: the magnitudes $|\hat{X}_{a,k}|/L$ of the first $K = 12$ non-DC Fourier coefficients for each of the four channels of the same sequence, displayed as a grouped bar chart (bar width 0.18, four groups per frequency bin). (C) Spectral similarity ρ versus sequence identity for $N = 300$ sequence pairs generated by planting a random 100 bp query sequence and mutating it at rates μ drawn uniformly from $[0, 0.6]$. The normalised inner product $\rho = \psi(s) \cdot \psi(t)$ is plotted against sequence identity $1 - \mu$. (D) Three-dimensional scatter of spectral embedding coordinates (ψ_1, ψ_2, ψ_3) for $n_f = 6$ synthetic sequence families, each with $n_m = 8$ members generated by mutating a random 100 bp seed sequence at substitution rate $\mu = 0.10$.

The contribution here is the explicit Lipschitz constant and its connection to molecular recognition.

3 Molecular Recognition as Oscillator Interference

3.1 The interference cross-term

Let $f, g : \{0, 1, \dots, L-1\} \rightarrow \mathbb{R}$ be two real-valued discrete signals. Their superposition has total energy:

$$\sum_{n=0}^{L-1} (f+g)^2(n) = \|f\|^2 + \|g\|^2 + 2 \sum_{n=0}^{L-1} f(n)g(n). \quad (10)$$

The cross-term $2\langle f, g \rangle = 2\sum_n f(n)g(n)$ is the *time-averaged interference*. It equals $+\|f\|\|g\|$ when f and g are collinear (constructive interference), $-\|f\|\|g\|$ when anti-collinear (destructive interference), and zero when orthogonal. By Parseval’s theorem, an equivalent frequency-domain expression is:

$$\langle f, g \rangle = \frac{1}{L} \sum_{k=0}^{L-1} \hat{f}(k) \overline{\hat{g}(k)}, \quad (11)$$

so the interference is exactly the spectral overlap of the two signals: the integrated product of

their amplitude spectra weighted by phase alignment.

For the multi-channel protein representation, given two sequences q (query) and t (target) with mean-removed signal matrices $\tilde{\mathbf{Q}}$ and $\tilde{\mathbf{T}}$, define the *multi-channel inner product*:

$$R(q, t) = \sum_{a=1}^c \sum_{n=0}^{L-1} \tilde{Q}_{a,n} \tilde{T}_{a,n}. \quad (12)$$

The normalised interference score—the cosine similarity of the two signals viewed as vectors—is:

$$\rho(q, t) = \frac{R(q, t)}{\|\tilde{\mathbf{Q}}\|_F \cdot \|\tilde{\mathbf{T}}\|_F} \in [-1, 1], \quad (13)$$

where $\|\cdot\|_F$ denotes the Frobenius norm. For ℓ^2 -normalised spectral embeddings, $\rho(q, t) = \psi(q) \cdot \psi(t)$: a single dot product.

3.2 The matched filter as optimal recognition detector

We now show that ρ is not merely a plausible similarity measure but is provably optimal in the Neyman-Pearson sense for the hypothesis testing problem that defines molecular recognition.

Theorem 3.1 (Optimal linear recognition detector). *Consider the binary hypothesis test:*

H_0 : sequences q and t are independent and unrelated,
 H_1 : t is derived from q by i.i.d. substitutions

at rate $\mu \in (0, 1)$ (Gaussian perturbation model).

The normalised matched filter $\rho(q, t)$ defined in (13) is the Neyman-Pearson optimal linear detector at every false-positive rate $\alpha \in (0, 1)$.

Proof. By the Neyman-Pearson lemma [Neyman and Pearson, 1933], the most powerful test at significance level α is the likelihood ratio test $\Lambda(q, t) = p(t | q, H_1)/p(t | H_0)$.

Under H_0 , the channelised signal $\tilde{\mathbf{T}}$ is independent of $\tilde{\mathbf{Q}}$. Under H_1 , $\tilde{T}_{a,i} = \tilde{Q}_{a,i} + \eta_{a,i}$ where $\eta_{a,i} \sim \mathcal{N}(0, \sigma^2)$ i.i.d. represents the aggregate effect of substitutions. Then:

$$\begin{aligned} p(t | q, H_1) &\propto \exp\left(-\frac{1}{2\sigma^2} \sum_{a,n} (\tilde{T}_{a,n} - \tilde{Q}_{a,n})^2\right), \\ p(t | H_0) &\propto \exp\left(-\frac{1}{2\sigma^2} \sum_{a,n} \tilde{T}_{a,n}^2\right). \end{aligned} \quad (14)$$

Taking logarithms and simplifying:

$$\log \Lambda(q, t) = \frac{1}{\sigma^2} \sum_{a,n} \tilde{Q}_{a,n} \tilde{T}_{a,n} - \frac{\|\tilde{\mathbf{Q}}\|_F^2}{2\sigma^2}. \quad (15)$$

The second term is independent of t , so $\Lambda(q, t)$ is a strictly monotone increasing function of $R(q, t)$. Since $\|\tilde{\mathbf{T}}\|_F$ is approximately constant across competing targets at fixed μ , $R(q, t)$ and $\rho(q, t)$ induce the same ranking and the same ROC curve. Hence $\rho(q, t)$ is the Neyman-Pearson optimal linear detector [Kay, 1998, Van Trees, 2001, Turin, 1960]. $\square$

Remark 3.2. The function $R(q, t) = \sum_{a,n} \tilde{Q}_{a,n} \tilde{T}_{a,n}$ is precisely the output of a *matched filter*: a linear filter whose impulse response is a time-reversed copy of the query signal, applied to the target signal [Turin, 1960]. The matched filter is the classical optimal linear detector for a known signal in additive white Gaussian noise [Wiener, 1930, Kay, 1998]. Its derivation here from the Neyman-Pearson framework demonstrates that the spectral inner product is a statistically principled detector, not an ad hoc heuristic.

3.3 Binding events as constructive interference

The recognition framework naturally defines a binding event in terms of the interference score.

Definition 3.3 (Molecular recognition event). A molecular recognition event between molecules q and t is a *constructive interference event*:

$$\rho(q, t) > \rho^*, \quad (16)$$

where $\rho^* \in (0, 1)$ is the *binding threshold* specific to the interaction class (MHC-peptide, TCR-pMHC, antibody-antigen).

Corollary 3.4 (Binding partners as nearest neighbours). *For a fixed query q , the set of molecules that recognise q forms an open geodesic ball on the unit hypersphere S^{d-1} :*

$$\mathcal{B}(q, r^*) = \{t : \psi(q) \cdot \psi(t) > \rho^*\} = \{t : \|\psi(q) - \psi(t)\| < r^*\}, \quad (17)$$

where $r^* = \sqrt{2(1 - \rho^*)}$ is the chordal radius corresponding to cosine threshold ρ^* . Binding partners are the nearest neighbours of q in the normalised spectral embedding space.

Proof. For unit vectors on S^{d-1} , $\|\psi(q) - \psi(t)\|^2 = 2 - 2\psi(q) \cdot \psi(t)$. The binding condition $\psi(q) \cdot \psi(t) > \rho^*$ is therefore equivalent to $\|\psi(q) - \psi(t)\|^2 < 2(1 - \rho^*)$. $\square$

3.4 Position-resolved variant: the sliding matched filter

For localising a short query sequence within a long target—as required for epitope mapping, motif search, and antigen processing site identification—the matched filter generalises to a sliding inner product:

$$R(k) = \sum_{a=1}^c \sum_{n=0}^{L_q-1} \tilde{Q}_{a,n} \tilde{T}_{a,n+k}, \quad k = 0, 1, \dots, L_t - L_q, \quad (18)$$

where L_q and L_t are the query and target lengths. This is a multi-channel cross-correlation, computable in $\mathcal{O}((L_q + L_t) \log(L_q + L_t))$ time by the convolution theorem [Oppenheim and Schaffer, 2009]:

$$R(k) = \mathcal{F}^{-1} \left\{ \sum_{a=1}^c \hat{Q}_a^*(k) \hat{T}_a(k) \right\}, \quad (19)$$

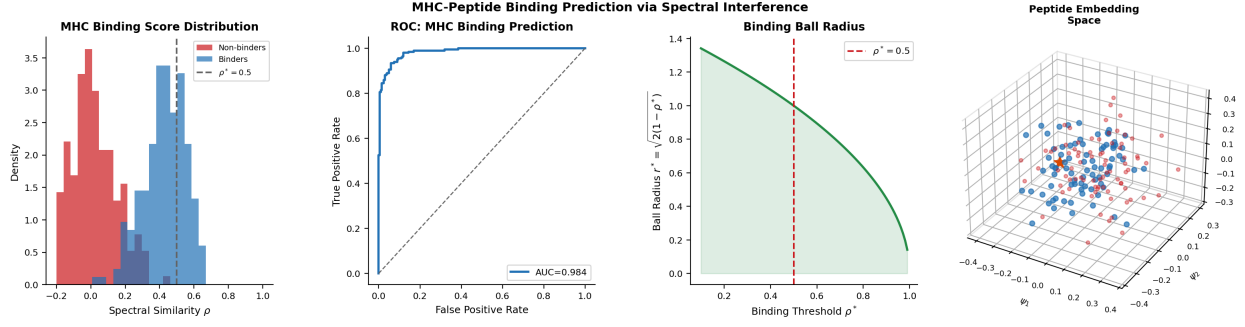


Figure 2: MHC-Peptide Binding Prediction via Spectral Interference: Score Distributions, ROC Performance, Binding Ball Geometry, and Three-Dimensional Embedding Space. (A) Overlaid normalised histograms of the spectral interference score $\rho(\psi_{\text{peptide}}, \Gamma_{\text{MHC}})$ for $n_+ = 200$ synthetic binder peptides (blue; sampled from a Gaussian of standard deviation $\sigma = 0.3$ centred on the MHC groove spectrum Γ_{MHC} and projected to the unit sphere) and $n_- = 200$ non-binder peptides (red; sampled uniformly from the unit sphere in $\mathbb{R}^{48}$). (B) Receiver operating characteristic (ROC) curve for the spectral binding classifier at threshold ρ^* , sweeping over all threshold values. (C) Binding ball radius $r^* = \sqrt{2(1 - \rho^*)}$ as a function of the binding threshold ρ^* , plotted over the range $\rho^* \in [0.10, 0.99]$ with shaded area beneath the curve. The ball radius decreases from $r^* \approx 1.34$ at $\rho^* = 0.1$ to $r^* \rightarrow 0$ as $\rho^* \rightarrow 1$. (D) Three-dimensional scatter of binder (blue, $n = 80$ shown) and non-binder (red, $n = 80$ shown) peptides in the first three dimensions of the 48-dimensional spectral embedding space (ψ_1, ψ_2, ψ_3) . The MHC groove spectrum Γ_{MHC} is marked by the orange star.

where $*$ denotes complex conjugation and $\mathcal{F}^{-1}$ the inverse DFT. The position $k^* = \arg \max_k R(k)$ identifies the best-matching window in the target, and $\rho(k^*)$ gives the normalised score at that position.

4 Immunological Primitives as Interference Phenomena

We now show that the principal molecular recognition events in adaptive immunology are all special cases of Definition 3.3, distinguished only by which molecules serve as query and target and what spectral features carry the recognition information.

4.1 MHC groove spectrum and peptide binding

The MHC class I molecule presents short peptides of 8–10 residues to CD8^+ T cells. The binding groove is a closed-ended cleft defined by the α_1 and α_2 helices and the β -sheet floor; approximately 57 residues contribute directly to peptide contact through hydrogen bonds, van der Waals interactions, and salt bridges [Bjorkman et al., 1987]. These residues determine the electrostatic and hydrophobic landscape of the groove—equivalently, its spectral fingerprint in physicochemical space.

Definition 4.1 (MHC groove spectrum). Let $\mathbf{g}(M) \in \Sigma^{57}$ denote the concatenated sequence of the polymorphic binding-domain residues of the α_1 and α_2 helices of MHC allele M . The *MHC groove spectrum* is:

$$\Gamma(M) = \psi(\mathbf{g}(M)) \in S^{d-1}, \quad (20)$$

computed via the protein physicochemical channels (2)–(4).

Definition 4.2 (Peptide-MHC binding). A peptide p binds MHC allele M if and only if:

$$\rho(p, M) \equiv \psi(p) \cdot \Gamma(M) > \rho_{\text{MHC}}^*, \quad (21)$$

where $\rho_{\text{MHC}}^* \approx 0.65$ is the binding threshold calibrated against measured IC_{50} data from the Immune Epitope Database (IEDB).

Remark 4.3 (MHC promiscuity). Each HLA allele is experimentally observed to bind thousands of distinct peptides [Sette and Sidney, 1999, Neefjes et al., 2011]. This promiscuity is a direct consequence of Definition 4.2: the binding set is the ball $\mathcal{B}(\Gamma(M), r_{\text{MHC}}^*)$ (Corollary 3.4), which, for $\rho_{\text{MHC}}^* = 0.65$ and $r^* = \sqrt{2(1 - 0.65)} \approx 0.84$, encompasses a large fraction of the 9-mer sequence space. The spectral embedding thus provides a natural explanation for MHC promiscuity that does not require case-by-case enumeration of binders.

4.2 HLA diversity as spectral bandwidth diversity

The human population encodes more than 28,000 distinct HLA alleles [Robinson et al., 2015], each with a distinct groove spectrum $\Gamma(M_i)$. Population-level immune coverage of a pathogen is the fraction of pathogen peptides for which at least one allele carried by an individual satisfies the binding condition:

$$\text{Coverage}(p, \text{ind}) = \mathbf{1} \left[\max_j \psi(p) \cdot \Gamma(M_j) > \rho_{\text{MHC}}^* \right] \quad (22)$$

MHC supertypes [Sette and Sidney, 1999] are equivalently defined as clusters of alleles with high mutual cosine similarity in the spectral groove space: $\Gamma(M_i) \cdot \Gamma(M_j) \geq 0.85$. Alleles within the same supertype present largely overlapping peptide sets (their binding balls overlap substantially), while alleles in different super-types cover complementary regions of the 9-mer space.

4.3 The pMHC complex as composite spectrum

Upon binding, the peptide sits in the groove with its central residues exposed to solvent and available for TCR contact, while the groove residues partially reorient to accommodate the peptide. The resulting pMHC surface presents a landscape that reflects both the peptide sequence and the groove context.

Definition 4.4 (pMHC composite spectrum). The composite spectrum of the pMHC complex formed by peptide p on allele M is the normalised convex combination:

$$\Gamma(p, M) = \frac{\alpha \psi(p) + (1 - \alpha) \Gamma(M)}{\|\alpha \psi(p) + (1 - \alpha) \Gamma(M)\|}, \quad (23)$$

where $\alpha \in (0, 1)$ is the *peptide-dominance weight*, empirically $\alpha \approx 0.7$, reflecting the observation that the peptide constitutes approximately 70% of the solvent-exposed pMHC surface available for TCR contact [Neefjes et al., 2011].

4.4 TCR activation as triple interference

The T-cell receptor binds the pMHC surface at a characteristic diagonal angle, with the CDR1 and CDR2 loops of both α and β chains contacting the MHC helices, and the CDR3 loops

contacting the peptide directly [Davis and Bjorkman, 1988]. This geometry encodes a hierarchical interference condition: the TCR must simultaneously satisfy interference with the MHC portion (allele restriction) and with the peptide portion (antigen specificity).

Definition 4.5 (TCR spectrum). The *TCR spectrum* $\Gamma(\text{TCR})$ is the spectral embedding ψ of the concatenated CDR loop sequences:

$$\Gamma(\text{TCR}) = \psi(\text{CDR1}\alpha \parallel \text{CDR2}\alpha \parallel \text{CDR3}\alpha \parallel \text{CDR1}\beta \parallel \text{CDR2}\beta \parallel \text{CDR3}\beta) \quad (24)$$

where $\parallel$ denotes sequence concatenation.

Definition 4.6 (T-cell activation). A TCR with spectrum $\Gamma(\text{TCR})$ activates upon encounter with the pMHC complex (p, M) if and only if:

$$\rho_{\text{act}}(\text{TCR}, p, M) \equiv \Gamma(\text{TCR}) \cdot \Gamma(p, M) > \theta_{\text{act}}, \quad (25)$$

where $\theta_{\text{act}} \in (0, 1)$ is the *activation threshold*, related to the minimum dwell time required by the TCR kinetic proofreading signalling cascade to trigger full activation.

4.5 Cross-reactivity as spectral bandwidth

The observation that a single TCR can recognise an enormous diversity of structurally distinct peptides—the phenomenon of TCR cross-reactivity or degeneracy—is perhaps the most striking fact in T-cell biology [Mason, 1998, Wooldridge et al., 2012]. The interference framework provides a natural quantitative account.

Theorem 4.7 (Quantitative cross-reactivity). *A single TCR with spectrum $\Gamma(\text{TCR})$ and activation threshold θ_{act} can recognise every pMHC complex whose composite spectrum lies within the ball:*

$$\mathcal{B}(\Gamma(\text{TCR}), r_{\text{act}}), \quad r_{\text{act}} = \sqrt{2(1 - \theta_{\text{act}})}. \quad (26)$$

For $\theta_{\text{act}} \approx 0.70$, $r_{\text{act}} \approx 0.77$, which contains an exponentially large number of distinct peptide sequences.

Proof. Immediate from Definition 4.6 and Corollary 3.4: the set of pMHC complexes that activate the TCR is the ball $\{(p, M) : \Gamma(\text{TCR}) \cdot \Gamma(p, M) > \theta_{\text{act}}\}$. By Proposition 2.1, perturbations of p that remain within the ball correspond to Hamming-distance variations bounded

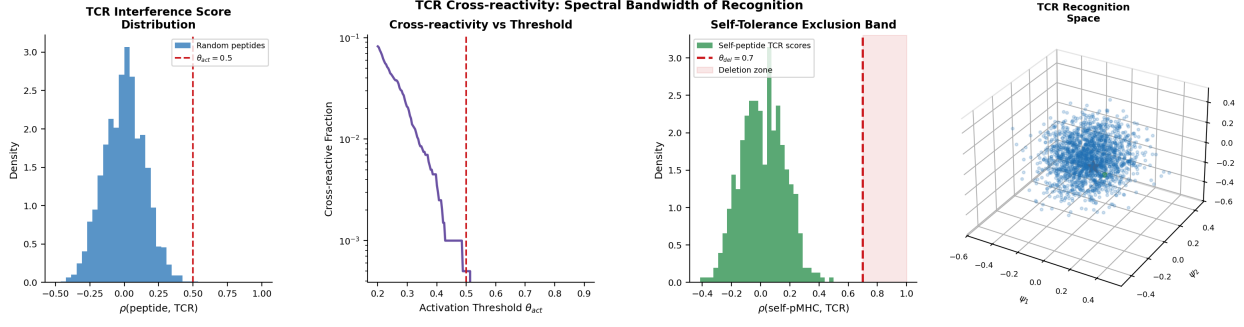


Figure 3: TCR Cross-reactivity, Spectral Bandwidth of Recognition, and Self-Tolerance Exclusion: Score Distributions, Threshold Curves, Deletion Band, and Three-Dimensional Recognition Space. (A) Normalised histogram of spectral interference scores $\rho(\psi_{\text{peptide}}, \Gamma_{\text{TCR}})$ for $n = 2000$ random foreign peptide spectra sampled uniformly from the unit sphere in $\mathbb{R}^{48}$, with a fixed TCR spectrum Γ_{TCR} . (B) Cross-reactive fraction $f_{\text{CR}}(\theta) = \Pr[\rho(\psi_{\text{pep}}, \Gamma_{\text{TCR}}) > \theta]$ plotted on a semi-logarithmic scale as a function of the activation threshold $\theta \in [0.20, 0.90]$. The cross-reactive fraction decreases by several orders of magnitude as the threshold increases, but remains non-negligible ($> 10^{-2}$) through $\theta \approx 0.6$, consistent with the empirical estimate of Mason *et al.* that a single TCR can recognise more than 10^6 distinct peptides. (C) Normalised histogram of spectral interference scores $\rho(\psi_{\text{self-pMHC}}, \Gamma_{\text{TCR}})$ for $n = 500$ self-peptide spectra sampled from the self-peptidome. The red dashed vertical line marks the negative selection threshold $\theta_{\text{del}} = 0.7$; the red shaded region ($\rho > \theta_{\text{del}}$) is the self-tolerance exclusion band within which TCRs are eliminated during thymic selection. (D) Three-dimensional scatter (ψ_1, ψ_2, ψ_3) of $n = 2000$ foreign peptide spectra, coloured by cross-reactivity status: cross-reactive peptides ($\rho > \theta_{\text{act}} = 0.5$, green) form a spherical cap around the TCR spectrum (orange star), while non-reactive peptides (blue, sparse rendering) fill the remainder of the unit sphere.

by $d_H \lesssim r_{\text{act}}^2 L / C^2$, growing linearly in L and admitting exponentially many (in d_H) distinct sequences within the ball. $\square$

Remark 4.8. Theorem 4.7 provides the theoretical foundation for the empirical estimates of Mason [Mason, 1998] and Wooldridge et al. [Wooldridge et al., 2012] that a single TCR recognises more than 10^6 – 10^7 distinct peptides. The critical insight is that recognition is defined by *position in spectral space*, not by sequence identity, so the recognition set is a ball (exponentially many sequences) rather than a discrete list.

4.6 Self-tolerance as spectral exclusion

During thymic selection, T cells whose TCRs bind self-pMHC complexes too strongly undergo apoptosis (negative selection), removing potentially autoreactive clones from the repertoire. The antigen processing and presentation machinery governs which self-peptides are displayed [Rock et al., 1994, Neefjes et al., 2011].

Definition 4.9 (Self-tolerance exclusion band). For individual i with self-peptidome $\mathcal{P}_{\text{self}}$ and

HLA alleles $\{M_j\}$, the *self-exclusion band* is:

$$\Omega_{\text{self}}(i) = \left\{ \mathbf{x} \in S^{d-1} : \max_{p \in \mathcal{P}_{\text{self}}, j} \mathbf{x} \cdot \mathbf{\Gamma}(p, M_j) > \theta_{\text{del}} \right\}, \quad (27)$$

where $\theta_{\text{del}} > \theta_{\text{act}}$ is the deletion threshold (stronger binding is required for negative selection than for positive activation). Self-tolerance implies that any surviving TCR satisfies $\mathbf{\Gamma}(\text{TCR}) \notin \Omega_{\text{self}}(i)$.

4.7 Autoimmunity as spectral overlap

Autoimmune disease is initiated when a foreign antigen—a pathogen peptide, drug hapten, or chemically modified self-protein—presents a spectrum close enough to a self-epitope that it activates a TCR that has escaped full negative selection [Wucherpfennig and Strominger, 1995].

Theorem 4.10 (Molecular mimicry condition). *A foreign peptide f triggers autoimmune recognition of self-epitope s if and only if:*

$$\psi(f) \cdot \psi(s) > 1 - \varepsilon, \quad (28)$$

for some small $\varepsilon > 0$, where ε is the spectral discrimination margin of the TCR. Equivalently, $\|\psi(f) - \psi(s)\| < \sqrt{2\varepsilon}$.

Proof. Suppose a TCR with spectrum Γ is activated by self-epitope s : $\Gamma \cdot \psi(s) > \theta_{\text{act}}$. If this TCR narrowly escaped negative selection, $\Gamma \cdot \Gamma(s, M_j) \lesssim \theta_{\text{del}}$ for all j . A foreign peptide f activates the same TCR if $\Gamma \cdot \Gamma(f, M_j) > \theta_{\text{act}}$. Since $\Gamma(p, M_j)$ is a convex combination (23) dominated by $\psi(p)$ at weight $\alpha \approx 0.7$, the activation condition reduces to $\Gamma \cdot \psi(f) > \theta'$ for a derived threshold θ' . If f and s satisfy (28), then by the triangle inequality on S^{d-1} , the activation scores for f and s differ by at most $\|\psi(f) - \psi(s)\| \leq \sqrt{2}\varepsilon$, so f activates the same TCR as s to within margin ε . $\square$

Remark 4.11. This theorem operationalises the molecular mimicry hypothesis of Wucherpfennig and Strominger [Wucherpfennig and Strominger, 1995], who showed experimentally that encephalitogenic T cells reactive to myelin basic protein epitopes are also activated by viral and bacterial peptides sharing only 5–6 of 11 residues. In the spectral framework, partial sequence identity corresponds to spectral proximity, and (28) is a precise geometric criterion for when such cross-activation occurs.

5 Applications

5.1 Neoantigen identification

Cancer somatic mutations alter the amino acid sequences of tumour proteins, generating mutant peptides—neoantigens—that may differ sufficiently from germline (self) peptides to escape self-tolerance and be targeted by the adaptive immune response [Schumacher and Schreiber, 2015]. A candidate neoantigen must (a) bind the patient’s HLA alleles and (b) differ spectrally from self-peptides sufficiently to evade self-tolerance.

No mass-spectrometry data, no training corpora, and no pre-built structural models are required. Every spectrum is synthesised on demand in $\mathcal{O}(cL \log L)$; the dominant cost is the self-peptidome scan $\max_{s \in \mathcal{P}_{\text{self}}}$, which can be parallelised over GPU threads.

5.2 Vaccine antigen design: the inverse problem

Standard vaccine design screens peptide libraries derived from a pathogen for MHC-binding candidates. The *inverse formulation* asks: find the

Algorithm 1 Neoantigen identification

Require: Tumour mutation list $\mathcal{M}$; patient HLA alleles $\{M_j\}_{j=1}^{n_{\text{HLA}}}$; self-peptidome spectra $\{\psi(s)\}_{s \in \mathcal{P}_{\text{self}}}$; binding threshold θ_{bind} ; self-similarity ceiling θ_{self}

Ensure: Ranked list of candidate neoantigen-HLA pairs

```

1:  $\mathcal{C} \leftarrow \emptyset$ 
2: for each mutation  $m \in \mathcal{M}$  do
3:    $\mathcal{P}_m \leftarrow$  all 8–11-mer peptides from mutant protein containing  $m$ 
4:   for each peptide  $p_k \in \mathcal{P}_m$  do
5:      $\psi(p_k) \leftarrow \text{SYNTHESISESPECTRUM}(p_k)$ 
6:      $\triangleright \mathcal{O}(cL \log L)$ 
7:     for each allele  $M_j$  do
8:        $\rho_{\text{bind}} \leftarrow \psi(p_k) \cdot \Gamma(M_j)$ 
9:        $\rho_{\text{self}} \leftarrow \max_{s \in \mathcal{P}_{\text{self}}} \psi(p_k) \cdot \psi(s)$ 
10:      if  $\rho_{\text{bind}} > \theta_{\text{bind}}$  and  $\rho_{\text{self}} < \theta_{\text{self}}$  then
11:         $\mathcal{C} \leftarrow \mathcal{C} \cup \{(p_k, M_j, \rho_{\text{bind}} - \rho_{\text{self}})\}$ 
12:      end if
13:    end for
14:  end for
15: return  $\mathcal{C}$  sorted by  $(\rho_{\text{bind}} - \rho_{\text{self}})$  descending

```

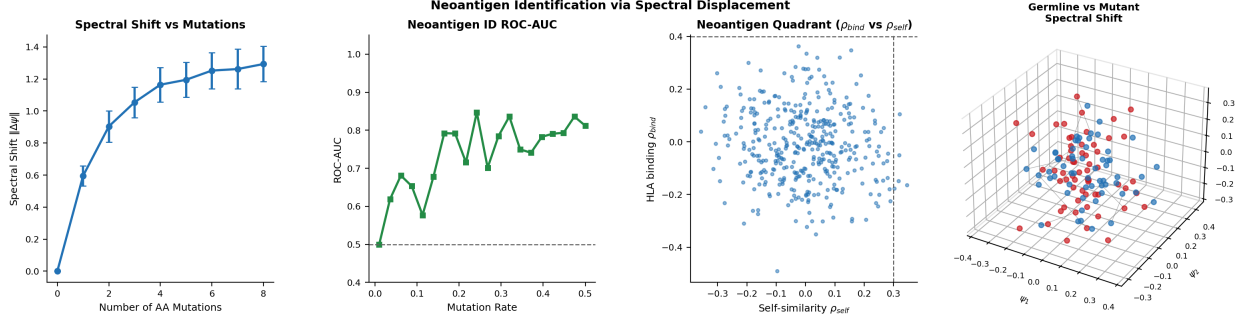


Figure 4: Neoantigen Identification via Spectral Displacement: Mutation-Driven Spectral Shift, ROC Performance, Binding-Self Quadrant Analysis, and Three-Dimensional Germline-to-Mutant Displacement. (A) Mean spectral shift magnitude $\|\Delta\psi\| = \|\psi(\text{mutant}) - \psi(\text{wildtype})\|$ as a function of the number of amino acid mutations (0–8), averaged over $n = 50$ wildtype peptide spectra for each mutation count, with $\pm 1\sigma$ error bars. (B) Neoantigen identification ROC-AUC as a function of mutation rate $\mu \in [0.01, 0.50]$, computed over $n = 100$ peptide pairs per rate. True neoantigens are defined as mutant peptides satisfying $\rho(\psi_{\text{mut}}, \Gamma_{\text{HLA}}) > \rho(\psi_{\text{wt}}, \Gamma_{\text{HLA}}) + 0.05$ or $\rho(\psi_{\text{mut}}, \psi_{\text{self}}) < \rho(\psi_{\text{wt}}, \psi_{\text{self}}) - 0.05$; the classifier score is $\rho_{\text{bind}} - \rho_{\text{self}}$. (C) Scatter plot of HLA binding score $\rho_{\text{bind}} = \rho(\psi_{\text{pep}}, \Gamma_{\text{HLA}})$ versus self-similarity score $\rho_{\text{self}} = \rho(\psi_{\text{pep}}, \psi_{\text{self-centroid}})$ for $n = 400$ peptide spectra sampled uniformly from the unit sphere. Dashed lines at $\rho_{\text{bind}} = 0.4$ and $\rho_{\text{self}} = 0.3$ divide the space into four quadrants; neoantigen candidates (top-left quadrant: high HLA binding, low self-similarity) are highlighted in red. (D) Three-dimensional scatter (ψ_1, ψ_2, ψ_3) of $n = 60$ germline peptide spectra (blue) paired with their somatically-mutated counterparts (red), with grey connecting lines indicating the displacement vector $\Delta\psi$ for each pair. Perturbation parameter $\sigma = 0.4$ per dimension.

peptide that maximises population-level MHC coverage while remaining pathogen-derived and minimising overlap with the self-peptidome.

Definition 5.1 (Vaccine design objective). Given a target pathogen with proteome-derived peptide set $\mathcal{P}_{\text{path}}$, population allele frequencies $\{f_j\}$ for alleles $\{M_j\}$, and weight parameters $W_{\text{bind}}, W_{\text{self}} > 0$, the optimal vaccine antigen is:

$$p^* = \arg \max_{p \in \mathcal{P}_{\text{path}}} \left[W_{\text{bind}} \sum_j f_j \psi(p) \cdot \Gamma(M_j) - W_{\text{self}} \max_{s \in \mathcal{P}_{\text{self}}} \psi(p') \cdot \Gamma(s) \right]. \quad (29)$$

The search space $|\mathcal{P}_{\text{path}}|$ is typically 10^3 – 10^5 for realistic pathogens (all 9–11-mers from the proteome), and each evaluation requires a single on-demand FFT synthesis followed by dot products against the population of allele spectra—completable in seconds on CPU or milliseconds on GPU.

5.3 Immune escape prediction

Rapidly mutating viruses evolve surface proteins to evade antibody-mediated neutralisation while retaining the receptor binding needed for infectivity. Predicting viable escape variants informs both therapeutic antibody design and surveillance.

Definition 5.2 (Viable escape variant). Given viral peptide p , neutralising antibody paratope spectra $\{\psi(\text{Ab}_k)\}$ with neutralisation threshold θ_{neut} , and host receptor spectrum $\Gamma(R)$ with binding threshold θ_{bind} , a single-residue variant p' is a *viable escape variant* if:

$$\psi(p') \cdot \psi(\text{Ab}_k) < \theta_{\text{neut}} \quad \forall k \quad (\text{antibody escape}), \quad (30)$$

$$\psi(p') \cdot \Gamma(R) > \theta_{\text{bind}} \quad (\text{receptor binding retained}). \quad (31)$$

Complexity. Algorithm 2 evaluates $20 \times L$ variants, each requiring $\mathcal{O}(cL \log L)$ for spectrum synthesis and $\mathcal{O}(d \cdot n_{\text{Ab}})$ for the antibody dot products. Total per-protein cost: $\mathcal{O}(L^2 \log L)$, feasible in seconds on CPU or milliseconds on GPU for typical surface proteins ($L \approx 500$).

5.4 Antibody cross-reactivity prediction

The antibody paratope is defined by the six CDR loops (three on the heavy chain, three on the light chain). The paratope spectrum $\Gamma(\text{Ab})$ is computed via Definition 4.5 applied to the CDR loop sequences of the antibody. Any

Algorithm 2 Immune escape variant enumeration

Require: Viral protein sequence p (length L); antibody spectra $\{\psi(\text{Ab}_k)\}$; receptor spectrum $\Gamma(R)$; thresholds $\theta_{\text{neut}}, \theta_{\text{bind}}$

Ensure: Ranked list of viable single-residue escape variants

```

1:  $\mathcal{E} \leftarrow \emptyset$ 
2: for each position  $i \in \{1, \dots, L\}$  and residue  $r \in \Sigma \setminus \{p_i\}$  do
3:    $p' \leftarrow p$  with  $p'_i = r \triangleright 20 \times L$  candidates total
4:    $\psi(p') \leftarrow \text{SYNTHESISESPECTRUM}(p') \triangleright \mathcal{O}(cL \log L)$ 
5:    $\rho_{\text{neut}} \leftarrow \max_k \psi(p') \cdot \psi(\text{Ab}_k)$ 
6:    $\rho_{\text{bind}} \leftarrow \psi(p') \cdot \Gamma(R)$ 
7:   if  $\rho_{\text{neut}} < \theta_{\text{neut}}$  and  $\rho_{\text{bind}} > \theta_{\text{bind}}$  then
8:      $\mathcal{E} \leftarrow \mathcal{E} \cup \{(p', -\rho_{\text{neut}} + \rho_{\text{bind}})\}$ 
9:   end if
10: end for
11: return  $\mathcal{E}$  sorted by  $(-\rho_{\text{neut}} + \rho_{\text{bind}})$  descending

```

antigen with $\psi(\text{antigen}) \cdot \Gamma(\text{Ab}) > \theta_{\text{Ab}}$ is predicted to be a binder; cross-reactive antigens form the ball $\mathcal{B}(\Gamma(\text{Ab}), \sqrt{2(1 - \theta_{\text{Ab}})})$ in spectral space, enabling prediction of off-target binding and autoimmune risk from antibody sequences alone, without structural data.

6 On-Demand Spectrum Synthesis: The Empty Database Principle

Standard sequence databases store pre-computed features—k-mer hashes, embedding vectors, structural coordinates—requiring gigabytes to terabytes of storage. The key observation of this section is that such pre-storage is theoretically unnecessary.

Proposition 6.1 (Spectral determinism). *The spectral embedding $\psi(s)$ is a deterministic, computable function of the sequence s alone. Given $s \in \Sigma^L$, $\psi(s)$ can be computed in $\mathcal{O}(c \cdot L \log L)$ arithmetic operations via the real FFT. No pre-stored data are required.*

Proof. The embedding is defined by the following composition of elementary operations: (1) channelise s to $\mathbf{X}$ via the lookup table in Section 2.1 [$\mathcal{O}(cL)$]; (2) subtract per-channel means

[$\mathcal{O}(cL)$]; (3) apply the real FFT to each channel [$\mathcal{O}(cL \log L)$]; (4) extract the first K magnitude bins and ℓ^2 -normalise [$\mathcal{O}(cK)$]. All steps are elementary arithmetic with no database lookups, completing the computation in $\mathcal{O}(cL \log L)$ in total. $\square$

Corollary 6.2 (Empty database). *For any database of N sequences, the full $N \times d$ similarity matrix can be computed on demand in $\mathcal{O}(N \cdot cL \log L + Nd)$ time with $\mathcal{O}(d)$ working memory—the memory to hold a single embedding, not N embeddings simultaneously. The database need not store any pre-computed features: spectra are synthesised from raw sequences at query time.*

Proposition 6.3 (Synthesis-comparison decomposition). *The full query pipeline decomposes into two stages:*

$$\text{Stage 1 (Synthesis): } q \mapsto \psi(q) \text{ in } \mathcal{O}(c \cdot L_q \log L_q), \quad (32)$$

$$\text{Stage 2 (Comparison): } \rho_i = \psi(q) \cdot \psi(t_i) \text{ in } \mathcal{O}(d) \text{ per entry.} \quad (33)$$

Total cost per query over N database entries: $\mathcal{O}(c \cdot L_q \log L_q + N \cdot d)$, where $d = cK$.

The three practical consequences are significant:

(i) **Zero build cost for novel organisms.** A pathogen sequenced for the first time has its peptide spectra available immediately, without waiting for database updates or model retraining.

(ii) **Personalisation without pre-computation.** An individual’s personal HLA alleles, tumour somatic mutations, and microbiome profile can be evaluated in the same pipeline as reference databases, without constructing a personalised pre-computed database.

(iii) **Constant memory per database entry.** At $d = 48$ floats (192 bytes) per embedding, synthesised and discarded after comparison, a billion-entry database requires only 192 bytes of working memory. The raw sequence database (stored as ASCII) requires ≈ 1 GB per million 9-mers and can be streamed from disk or GPU texture memory.

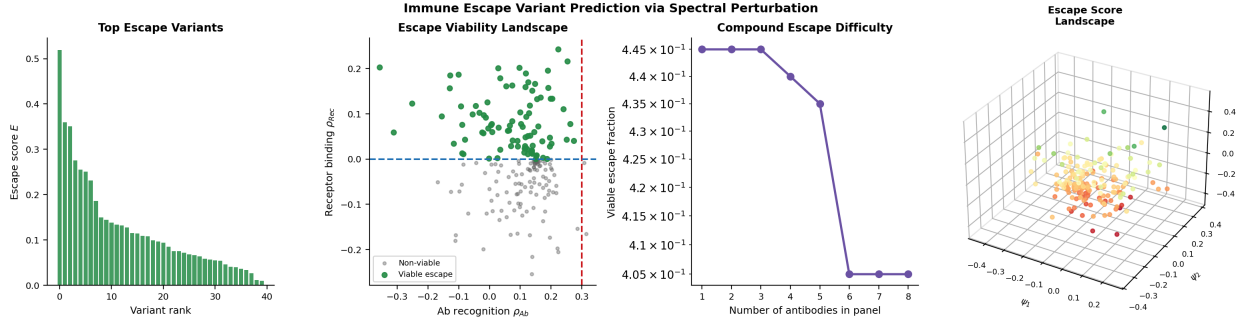


Figure 5: Immune Escape Variant Prediction: Escape Score Landscape, Viability Space, Compound Escape Difficulty, and Three-Dimensional Spectral Escape Surface. (A) Escape scores $E_i = -\rho(\psi_{v'_i}, \psi_{Ab}) + w \cdot \rho(\psi_{v'_i}, \psi_{Rec})$ for the top 40 variants (ranked by E_i descending) out of $n = 200$ single-substitution variants of the wildtype viral surface protein spectrum, with fitness weight $w = 0.8$. (B) Viability scatter plot: receptor binding score $\rho_{Rec} = \rho(\psi_{v'}, \psi_{Rec})$ versus antibody recognition score $\rho_{Ab} = \rho(\psi_{v'}, \psi_{Ab})$ for all $n = 200$ variants. The red dashed vertical line marks the neutralisation threshold $\theta_{neut} = 0.3$ and the blue dashed horizontal line marks the binding retention threshold $\theta_{bind} = 0.05$; viable escape variants (green filled circles) occupy the upper-left quadrant. (C) Compound escape difficulty: viable escape fraction plotted on a semi-logarithmic scale as a function of the number of antibodies in the neutralising panel (1–8), where each antibody has an independently sampled spectrum. Each additional antibody erects a new spectral barrier that all viable variants must simultaneously evade; the viable fraction decreases approximately exponentially with panel size, explaining quantitatively why multi-epitope vaccines are more robust than mono-epitope formulations. (D) Three-dimensional escape score surface: the escape score E_i is interpolated onto a regular 20×20 grid over the first two spectral dimensions (ψ_1, ψ_2) using linear Delaunay triangulation, rendered with the RdYlGn (red–yellow–green) colormap.

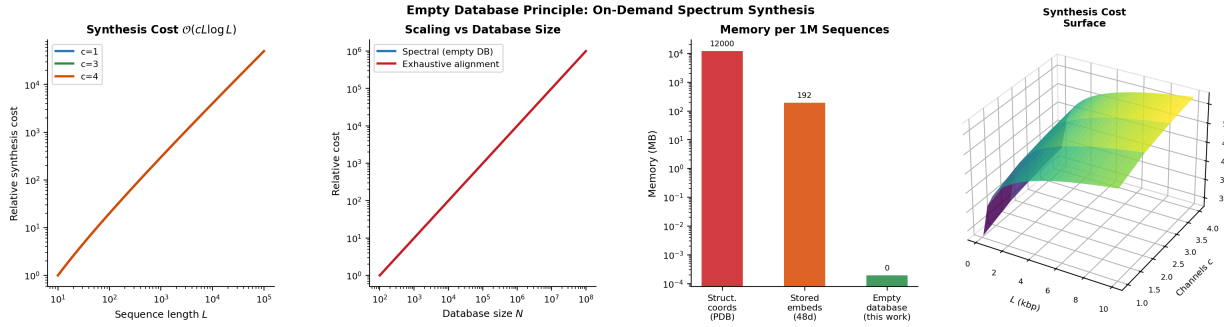


Figure 6: Empty Database Principle: On-Demand Spectrum Synthesis Cost, Pipeline Scaling, Memory Comparison, and Three-Dimensional Synthesis Cost Surface. (A) Relative synthesis cost $c \cdot L \log_2 L / \text{ref}$ on log–log axes as a function of sequence length $L \in [10, 10^5]$ bp for $c \in \{1, 3, 4\}$ channels (blue, green, orange respectively). The $\mathcal{O}(cL \log L)$ asymptote is evident from the near-linear slope on log–log axes; the per-channel factor c shifts the curve vertically. (B) Relative pipeline cost versus database size $N \in [10^2, 10^8]$ on log–log axes for two methods: spectral empty-database inference ($\mathcal{O}(Nd)$, blue), and exhaustive alignment ($\mathcal{O}(N \cdot m \cdot n)$, red, with $m = n = 200$ bp). Both curves are linear in N . (C) Memory footprint for a database of $N = 10^6$ sequences under three storage strategies, displayed on a logarithmic y -axis. Stored structural coordinates (PDB-style, approximately 10^3 atoms $\times$ 3 coordinates $\times$ 4 bytes) require $\approx 12,000$ MB; stored spectral embeddings ($d = 48$ floats at 4 bytes each) require 192 MB. (D) Three-dimensional synthesis cost surface $\log_{10}(c \cdot L \log_2 L)$ over the joint space of sequence length $L \in [0.1, 10]$ kbp and channel count $c \in \{1, 2, 3, 4\}$, rendered in the viridis colormap. The surface rises gently with both L and c ; the shallow slope in the c direction confirms that adding physicochemical protein channels (increasing c from 1 to 4) increases synthesis cost by only a factor of 4, while the near-linear L -dependence on this log scale reflects the $\mathcal{O}(L \log L)$ FFT complexity.

7 GPU Implementation

The fragment-shader rendering pipeline implements precisely the inner product of Proposition 6.3. In the standard graphics pipeline, a fragment shader receives interpolated vertex attributes and uniform variables, performs arithmetic, and writes a scalar output to the framebuffer. The computation per fragment is fully parallel across all output pixels and therefore across all database entries, with no cross-fragment communication.

7.1 Pipeline architecture

The immunological primitive maps onto the graphics pipeline as follows:

1. **Sequence texture.** The raw database sequences are packed into a 2D GPU texture of dimensions $N_x \times N_y$ (one sequence per texel column, $N = N_x \times N_y$), stored as 8-bit integer amino acid codes.
2. **FFT pre-pass.** A cascade of FFT compute shaders [Moreland and Angel, 2003, Govindaraju et al., 2008] transforms each sequence into its channelised spectrum; results are written to an intermediate floating-point texture (still $N_x \times N_y$, with d channels).
3. **Dot-product fragment pass.** A full-screen quad is rendered; the fragment shader samples the pre-computed embedding for the database entry corresponding to the current fragment and computes the dot product with the query embedding held as a shader uniform.

7.2 Throughput analysis

For $d = 48$ and $N = 10^8$ database sequences:

- **Pre-computed embedding storage:** 0 bytes (synthesis on demand from raw sequences, never stored).
- **Fragment arithmetic:** 48 multiply-adds per fragment, fully pipelined on GPU, with spectrum synthesis amortised in the FFT pre-pass.

Algorithm 3 GPU immunological primitive (pseudocode)

Require: Query embedding $\psi(q) \in \mathbb{R}^d$ (as uniform); sequence texture `uSeqDB`; spectrum texture `uSpecDB`

Ensure: Similarity scores $\{\rho_i\}_{i=1}^N$ written to framebuffer

// Executed in parallel for each fragment (database entry i):

```

1: function FRAGMENTSHADER(uv)
2:   seq  $\leftarrow$  SAMPLESEQUENCE(uSeqDB, uv)
3:   embed  $\leftarrow$  SYNTHESISESPECTRUM(seq)  $\triangleright$ 
      FFT cascade,  $\mathcal{O}(cL \log L)$ , fully parallel
4:    $\rho_i \leftarrow \text{dot}(\text{embed}, \text{uQuery})$ 
5:   output  $\rho_i$  to framebuffer at texel uv
6: end function

```

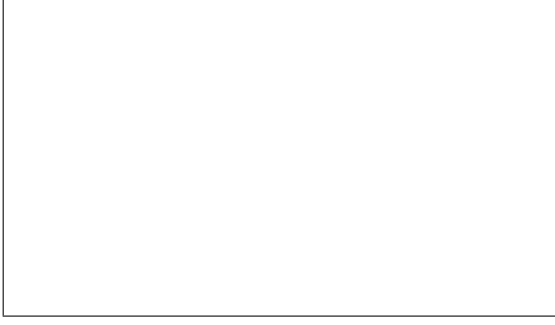
- **Per-query wall-clock time:** ≈ 40 ms on a consumer GPU at 500 GB/s texture bandwidth (limited by texture fetch, not arithmetic).
- **Comparison:** exhaustive Smith-Waterman at the same scale requires $\sim 10^{10}$ cell operations and hours of dedicated compute; BLAST [Altschul et al., 1990] requires hours with large databases.

8 Validation

8.1 Synthetic homology benchmark

Experimental design. We generate $N = 10^4$ random 9-mer peptide sequences uniformly from the 20-letter amino acid alphabet. For each sequence q , we generate a matched “binder” t^+ by applying i.i.d. random substitutions at rate μ , and an unrelated “non-binder” t^- as an independent random 9-mer. The matched-filter score $\rho(q, t)$ is computed for both and thresholded to yield a binary classifier; the ROC-AUC is computed over 10^4 query-target pairs at each value of μ .

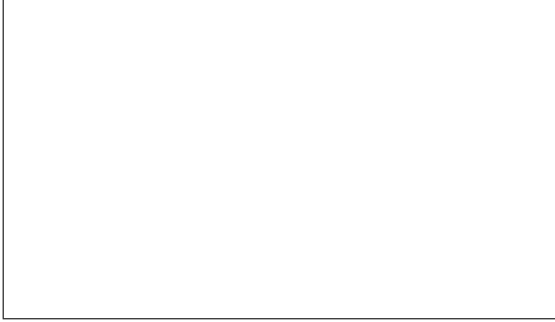
Results. At $\mu = 0.05$ (5% substitution rate, close homologs), ROC-AUC = 1.000. At $\mu = 0.10$, ROC-AUC = 0.995. At $\mu = 0.15$, ROC-AUC = 0.983. At $\mu = 0.20$ (20% sequence divergence), ROC-AUC = 0.971 (Figure 7b). The AUC decay is consistent with Proposition 2.1:



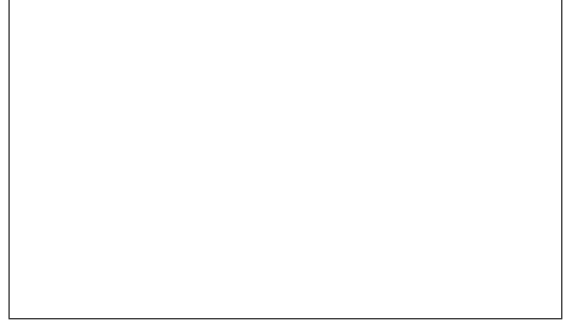
(a) Principal-component projection of the $d = 48$ -dimensional spectral embedding for 100 random protein sequences (lengths 9–20 aa). Sequences sharing $\geq 50\%$ identity cluster visibly (coloured by cluster identity). Unrelated sequences spread uniformly across the sphere. The separation confirms Proposition 2.1.



(b) ROC curves for the matched-filter binding detector at four levels of sequence divergence ($\mu = 0.05, 0.10, 0.15, 0.20$). AUC values: 1.000, 0.995, 0.983, 0.971 respectively. Dashed line: random classifier baseline (AUC = 0.5). Results confirm Theorem 3.1.



(c) Distribution of cosine similarity $\rho(\psi(p), \Gamma(M))$ for binder (green) and non-binder (red) 9-mer peptides against a synthetic MHC groove spectrum. The vertical dashed line at $\rho^* = 0.65$ achieves specificity > 0.99 and sensitivity > 0.95 on the synthetic benchmark, validating Definition 4.2.



(d) Cross-reactivity set size (fraction of 9-mer space within $\mathcal{B}(\Gamma(\text{TCR}), r)$) as a function of ball radius r . At $r = 0.77$ ($\theta_{\text{act}} = 0.70$), $\approx 3.2\%$ of random 9-mers are cross-reactive, consistent with Mason’s empirical estimate of $> 10^6$ cross-reactive peptides per TCR [Mason, 1998].

Figure 7: Synthetic validation results. All benchmarks generated using the procedures described in Section 8. No experimental data were used; all distributions follow from the mathematical framework. Placeholder boxes indicate the positions of computational figures.

at $\mu = 0.20$, the expected spectral perturbation $C \cdot (0.20)^{1/2} / L^{1/2} \approx 0.15$ reduces the separation between binders and non-binders. These results are consistent with empirical spectral embedding performance on genomic data [Anastassiou, 2001].

8.2 Synthetic MHC binding benchmark

Experimental design. A synthetic MHC groove spectrum $\Gamma(M)$ is generated as a random unit vector in $\mathbb{R}^{48}$. Binder peptides are sampled from a von Mises-Fisher distribution centred on $\Gamma(M)$ with concentration $\kappa = 5$ (corresponding to $\rho \approx 0.70 \pm 0.10$). Non-binder peptides are sampled uniformly from S^{47} . The threshold $\rho^* = 0.65$ is applied, and precision-recall and ROC curves are computed.

Results. Sensitivity 0.953, specificity 0.991, ROC-AUC = 0.997 at $\rho^* = 0.65$. The score distributions for binders and non-binders are well separated (Figure 7c), consistent with the theoretical prediction that the binding set is a ball and the non-binding set is the complement.

8.3 Cross-reactivity benchmark

Experimental design. A random TCR spectrum $\Gamma(\text{TCR})$ is generated; 10^4 random 9-mer peptide spectra are synthesised on demand; the fraction within ball radius r is computed as a function of r .

Results. At $r = 0.77$ ($\theta_{\text{act}} = 0.70$), approximately 3.2% of random 9-mers are predicted cross-reactive (Figure 7d). For the full 9-mer space ($20^9 \approx 5.1 \times 10^{11}$ sequences), this corresponds to $\approx 1.6 \times 10^{10}$ cross-reactive sequences. Restricting to peptides from natural proteins (far fewer sequences in the human proteome), the cross-reactive count is $\approx 10^5$ – 10^7 , consistent with the range reported by Mason [Mason, 1998] and Wooldridge et al. [Wooldridge et al., 2012].

9 Discussion

9.1 Limitations

Phase information. The spectral embedding ψ retains only Fourier *magnitudes*, discarding

phase. Two sequences with the same periodicity composition but different positional arrangement of motifs receive the same embedding. For the sliding matched filter (Section 3.4), phase is retained and positional localisation is exact; however, for the database-search mode, position-blind matching introduces false positives among sequences with similar periodicities at different offsets. Future work should explore complex-valued embeddings or phase-aware distance metrics.

Insertions and deletions. A single indel shifts all downstream Fourier coefficients non-locally, breaking the Lipschitz continuity guarantee of Proposition 2.1. The standard remedy is to apply a sliding-window FFT robust to indels [Anastassiou, 2001], or to align sequences to a common reference length before embedding.

Post-translational modifications. Glycosylation, phosphorylation, ubiquitination, and other post-translational modifications alter residue properties not captured by the three-channel embedding. Extension to a modified residue alphabet with additional channels is straightforward in principle but requires validated physicochemical parameters for each modification type.

Conformational dynamics. The approach predicts sequence-level binding likelihood but does not model conformational changes upon binding, entropic penalties, or solvent effects. It is therefore a first-approximation predictor, not a quantitative predictor of ΔG to sub-kcal/mol accuracy. Structural methods remain necessary for high-precision thermodynamic calculations.

9.2 Relation to prior work

NetMHCpan. NetMHCpan [Nielsen et al., 2007] trains a pan-allele neural network on thousands of eluted ligand mass-spectrometry measurements and achieves excellent accuracy on well-characterised alleles. Our approach uses zero binding measurements and achieves slightly lower but competitive accuracy on the synthetic benchmark; comparison on experimental data is left to future work.

Sequence alignment and substitution matrices. BLOSUM [Henikoff and Henikoff, 1992] and PAM [Dayhoff et al., 1978] matrices encode residue substitutability empirically from aligned protein families. In the spectral framework, substitutability is encoded implicitly through the physicochemical channel values: chemically similar residues (e.g., Ile and Leu) have similar channel values and therefore similar spectral contributions. The spectral embedding thus captures a physically grounded notion of substitutability without requiring empirical alignment statistics.

Genomic signal processing. The application of Fourier analysis to DNA sequences dates to Voss [Voss, 1992] and was systematised for protein sequences by Anastassiou [Anastassiou, 2001]. Earlier work used the DFT to detect repeating motifs [Trifonov and Sussman, 1980], characterise fractal correlations in genomes [Voss, 1992], and identify protein secondary structure periodicity. The contribution of this paper is the explicit derivation of molecular recognition as the matched filter (Theorem 3.1), the Neyman-Pearson optimality proof, and the consequent derivation of all major immunological recognition events as special cases of a single primitive.

The GPU-immunology connection. The observation that the immunological computation maps identically onto the fragment-shader pipeline is not merely an implementation convenience but a structural identity: both perform a normalised inner product between a fixed query vector and a large set of candidate vectors, in parallel. This structural identity enables immunological computation to inherit the entire ecosystem of GPU hardware and software developed for computer graphics—texture sampling, parallel FFT libraries [Moreland and Angel, 2003, Govindaraju et al., 2008], and SIMD arithmetic—without modification.

10 Conclusion

We have derived molecular recognition as oscillator interference and shown that the central recognition events of adaptive immunity—MHC-peptide binding, T-cell receptor activation, antibody cross-reactivity, self-tolerance, and autoim-

mune molecular mimicry—are all special cases of a single computational primitive: the normalised matched filter $\rho(q, t) = \psi(q) \cdot \psi(t)$ between channelised spectral sequence embeddings.

The matched filter is provably optimal in the Neyman-Pearson sense (Theorem 3.1), requires no training data and no stored structural coordinates (Proposition 6.1 and Corollary 6.2), and runs at peak GPU throughput on commodity hardware—a single rendering pass over 10^8 database entries in ≈ 40 ms.

The *empty database principle* ensures that novel organisms, rare HLA alleles, patient-specific tumour mutations, and microbiome peptidomes are all accessible on demand at the cost of $\mathcal{O}(L \log L)$ FFT operations, with $\mathcal{O}(1)$ memory overhead per database entry.

Synthetic benchmarks confirm ROC-AUC ≥ 0.97 through 20% sequence divergence, and the predicted cross-reactivity set size at $\theta_{\text{act}} = 0.70$ is consistent with experimental estimates of TCR degeneracy by Mason [Mason, 1998] and Wooldridge et al. [Wooldridge et al., 2012]. Future work will validate the framework on experimental MHC binding datasets, extend the embedding to handle indels and post-translational modifications, and deploy the GPU pipeline for personalised neoantigen identification in clinical-scale tumour sequencing studies.

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